Predicting Diabetes Status from Population Health Indicators Using Traditional Machine Learning and Deep Learning

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GitHub repository: `https://github.com/ndunge-mbithi/summative_intro-_to_ML`
Google Drive folder: *[link pending — folder not yet created]*
Video presentation: *[link pending — video not yet recorded]*

***Abstract*—**Motivated by the large share of undiagnosed diabetes in rural sub-Saharan Africa, where questionnaire-based triage is far more scalable than laboratory testing, I compare traditional machine learning and deep learning for classifying self-reported diabetes status (no diabetes, prediabetes, diabetes) from 25 population health indicators in the raw 2015 CDC Behavioral Risk Factor Surveillance System. The data comprise 441,456 records, verified against the official codebook, of which 440,658 have a valid label. Using a leakage-safe pipeline that keeps respondents with missing predictors, I ran 17 tracked experiments covering logistic regression, random forests, gradient boosting, and TensorFlow/Keras networks built with the Sequential and Functional APIs and fed through `tf.data`. The experiments varied hyper-parameters, engineered features, class weighting and regularisation. On a held-out test set of 66,099 respondents, tuned gradient boosting achieved a macro ROC-AUC of 0.800 and a macro average precision of 0.479. A wide-and-deep network matched it (0.800 and 0.478), and logistic regression followed closely (0.794 and 0.469). Diabetes was moderately well discriminated (ROC-AUC 0.843), but prediabetes was not (average precision 0.036). Decision rules mattered more than algorithms: a validation-selected F_2 threshold raised diabetes sensitivity from 0.17 to 0.85, at the cost of flagging 39% of respondents. Error, explainability and subgroup analyses point to label limitations, reverse causation and uneven sensitivity across age, income and racial groups. The models are research tools, not diagnostic systems.

***Index Terms*—**diabetes, BRFSS, class imbalance, gradient boosting, deep learning, population health, screening.

1. Introduction

Diabetes now affects roughly one adult in ten. The International Diabetes Federation estimated that 536.6 million people aged 20–79, or 10.5% of that age group, were living with the disease in 2021, and it expects the figure to reach 783.2 million (12.2%) by 2045 [1]. Worse, many people with type 2 diabetes and most with prediabetes do not know they have it, so complications build unnoticed and much effort goes into deciding who should be tested first.

This question matters to me because of the work I want to do. My mission is to revolutionise healthcare in sub-Saharan Africa through technology, creating accessible, equitable and scalable solutions for underserved rural communities, and it is in exactly those communities that the diagnosis gap is widest. The IDF reports that 24 million African adults had diabetes in 2021 and that 54% of them were undiagnosed [2]. Kenya's 2015 national STEPS survey found that more than half of adults with diabetes were unaware of their condition, and measured prevalence was lower in rural areas [3], where fewer people are ever tested. In a rural clinic without a laboratory, a short questionnaire administered by a community health worker is about the most scalable screening tool available. Before anyone builds such a tool, though, someone has to establish how much a questionnaire can actually reveal, and what it costs in missed cases and false alarms. That is the gap this project tries to fill.

I could not do this with African data. No openly available survey from the region yet combines a large sample with richly documented predictors; the Kenyan STEPS analysis, for example, covered 4,069 adults [3]. Instead I use the US Behavioral Risk Factor Surveillance System (BRFSS), which interviews hundreds of thousands of adults each year about chronic conditions, behaviour, access to care and demographics [4], and whose 2015 file is documented variable by variable in its codebook [5]. I treat it as a testbed. Whatever it teaches has to be revalidated on local data before it is used anywhere in Africa. The BRFSS has also attracted a growing machine learning literature, but as Section 2 shows, much of that work simplifies the outcome to a binary label, uses a cleaned subset of the data, and reports accuracy at default thresholds, choices that make the task look easier than it really is.

My research question is therefore: *How effectively can traditional machine learning and deep learning models classify diabetes status from population health indicators, and how do their predictive performance, interpretability, computational efficiency and error patterns differ?* To answer it, I kept the survey's own three categories of no diabetes, prediabetes and diabetes, built a leakage-safe pipeline, and compared logistic regression, random forests and gradient boosting with Sequential and Functional TensorFlow/Keras networks across 17 tracked experiments, before examining the selected models' thresholds, errors, explanations and subgroup performance.

Going in, I expected gradient boosting to give the best discrimination, neural networks to match it without beating it, and logistic regression to trail only slightly. Benchmark studies point that way for tabular data [6, 7], clinical reviews find little advantage for complex models [8], and most survey answers are simple yes/no or ordered categories whose effects are likely to be additive. I also expected prediabetes to be very hard to separate from its neighbours. Throughout, I keep in

mind that the labels are self-reported answers to the question of whether a doctor ever said the person had diabetes. They are not laboratory results, and none of the models here is a diagnostic tool.

2. Literature Review

2.1. Classifying diabetes from survey data

The study closest to mine is by Xie *et al.* [9], who built type 2 diabetes risk models from the 2014 BRFSS. Their pipeline narrowed the data considerably: of 464,644 respondents, 138,146 remained after preprocessing, the outcome was restricted to non-pregnant adults over 30, and people with prediabetes were left out altogether. They compared eight classifiers on a two-thirds/one-third holdout and applied SMOTE only to the training portion, which is the correct way to resample. Discrimination was modest and tightly bunched, with AUCs between 0.718 and 0.795. The neural network scored best on AUC and accuracy yet caught only 37.8% of cases, so the authors recommended the decision tree for screening because it found more of them. The paper taught me to judge screening models by sensitivity, and showed how much a filtered binary outcome can flatter results. One reported "risk factor" also stayed with me: a recent check-up was linked to *higher* risk, almost certainly because people with diabetes see doctors more often. Survey predictors can reflect a diagnosis as much as anticipate it.

Most recent work uses a cleaned derivative of the 2015 BRFSS with 253,680 rows and 21 predictors, distributed through Kaggle and the UCI repository. Majcherek *et al.* [10] compared 18 algorithms on it and reported that an Extra Trees model reached an AUC of 0.99 and accuracy above 90%, with BMI, age, general health and income as leading SHAP features. Zafar *et al.* [11] applied ADASYN oversampling to the same data and proposed a hybrid deep architecture explained with LIME and SHAP. These results are hard to reconcile with the audit by Chowdhury *et al.* [12], which found AUROCs of only 0.818 to 0.830 across four learners, with gradient boosting already within 0.0003 of the estimated ceiling for what the survey variables can support. An AUC of 0.99 on data with such a ceiling suggests that something in the evaluation, perhaps oversampling or duplicate rows crossing the train–test boundary, deserves a closer look. I cannot tell the cause from the published papers, so I treat those figures as reported claims rather than benchmarks. The derivative dataset also makes two choices I wanted to avoid. Its binary version counts prediabetes as diabetes, which changes the meaning of the outcome, and it drops every respondent with a missing answer, even though people who refuse questions or stop interviews early are not a random subset.

2.2. Simple models, complex models and tabular data

Whether flexible models beat simple ones is an old question in clinical prediction, and the evidence is sobering. Christodoulou *et al.* [8] reviewed 71 studies containing 282 comparisons and found no difference in discrimination between logistic regression and machine learning among comparisons at low risk of bias (difference in logit AUC 0.00, 95% CI -0.18 to 0.18). The apparent advantages of machine learning were concentrated in the studies with weaker designs. Kopitar *et al.* [13] similarly found no clinically relevant gain from tree ensembles over regression for early type 2 diabetes detection.

The machine learning benchmarks tell a compatible story about neural networks. Grinsztajn *et al.* [6] tested deep architectures against tree ensembles on 45 datasets and found that trees remained state of the art on medium-sized tabular data. They traced the gap to inductive biases: trees cope well with uninformative features, respect the orientation of the feature axes and fit irregular functions easily. Shwartz-Ziv and Armon [7] found that XGBoost beat recently proposed deep tabular models on most of the very datasets those models had been introduced with, and needed less tuning to do so. Whether this holds for health surveys, where nearly every predictor is binary or ordinal, is something I test directly. The ensembles I compare descend from Friedman's gradient boosting machine [14] and Breiman's random forests [15], and my Functional API network adapts the wide-and-deep design of Cheng *et al.* [16], which places a linear model alongside a deep one.

2.3. Evaluating rare outcomes

Diabetes is a minority class in any general population survey, and prediabetes is rarer still. Saito and Rehmsmeier [17] showed that ROC curves can look reassuring in this situation because the false-positive rate is computed against a large pool of negatives, while precision–recall curves, whose baseline equals the prevalence, show how hard the problem really is. That argument shaped my choice of average precision as the main metric. Resampling only helps when confined to training data, and published work is not always clear about this; the TRIPOD+AI statement [18] now asks authors to report such choices along with missing-data handling and outcome definitions.

2.4. Explanations, fairness and the quality of labels

Impurity-based importances from random forests are biased towards variables with many categories or continuous scales [19], so I rely instead on permutation importance and SHAP [20], which divides each prediction among the features using Shapley values. Fairness raises a deeper problem. Obermeyer *et al.* [21] showed that a widely used population-health algorithm under-identified Black patients because it predicted health-care cost as a stand-in for health need. A similar trap exists here. A self-reported "ever told" diagnosis depends on access to care, so the groups who see doctors least are also the ones most likely to be labelled wrongly. Finally, a systematic review found that BRFSS self-reports agree reasonably with other self-report surveys but less well with surveys that take physical measurements [22], which limits how far any model trained on these labels can be read clinically.

2.5. The gap this project addresses

Taken together, these studies leave three gaps. Machine learning work on BRFSS diabetes data usually models either a filtered binary outcome [9] or the complete-case derivative

Table 1: DIABETE3 categories, observed counts (identical to the codebook [5]) and the 3-class target mapping.

Code	Response	*n*	Target class
1	Yes	57,256	Diabetes
2	Yes, only during pregnancy	3,608	No diabetes
3	No	372,104	No diabetes
4	No, pre-/borderline diabetes	7,690	Prediabetes
7	Don't know	598	removed
9	Refused	193	removed
missing	Blank	7	removed
	Total	441,456	

with prediabetes merged into diabetes [10, 11], so the three categories the survey actually records are rarely modelled. Evaluations tend to stop at accuracy and ROC-AUC under default thresholds, saying little about minority-class precision, threshold choice or differences between groups of people. And comparisons of traditional and deep models seldom ask whether the extra complexity buys anything over a well-tuned linear model trained on the same, leakage-safe pipeline. My project works on the raw 2015 survey, keeps prediabetes as its own class, retains respondents with missing answers, and compares both paradigms under identical preprocessing with tuned decision rules and subgroup analysis. Individual pieces of this have been done before; the contribution lies in putting them together in one transparent and reproducible comparison.

3. Methodology

3.1. Data, provenance and target

I worked with the 2015 BRFSS combined landline and cellphone public-use file [4], which holds 441,456 interviews with non-institutionalised US adults across 330 variables. My sandboxed environment could not reach the CDC website, so I took a copy of `LLCP2015.XPT` from a public research repository. Because it came from a mirror, the loader checks its record count, variable count and every target category against the official codebook [5] and refuses to continue on any mismatch. Every count matched exactly (Table 1); in Colab the same loader downloads the file from the CDC.

The target, `DIABETE3`, records the answer to "(Ever told) you have diabetes". I grouped its answers into three classes: no diabetes (codes 3 and 2), prediabetes (code 4) and diabetes (code 1). Code 2, diabetes only during pregnancy, describes a past condition, so following CDC's convention I counted it as no diabetes. Removing the 798 "don't know", "refused" and blank answers left 440,658 respondents. I kept prediabetes separate, since merging it with diabetes would change the outcome's meaning and hide the hardest part of the problem.

3.2. Choosing and recoding the predictors

I chose 25 core-questionnaire predictors linked to diabetes in the literature, covering cardiometabolic and general health, behaviour and BMI, and demographics and access to care (the full list is in the repository). I left out the optional diabetes modules, whose questions, such as insulin use, are only asked of people who already report diabetes and would leak the answer into the inputs. Race and ethnicity were kept aside for checking performance across groups and never used as model inputs.

Every recode follows the codebook and is unit-tested: "don't know" and "refused" codes became missing values rather than numbers, 88 ("no days") became zero, and BMI was rescaled from its implied decimals. The cholesterol question needed more thought. It is only asked of people who have been tested, so 49,673 never-tested respondents appear blank, yet someone never tested has, by definition, never been told their cholesterol is high. I coded them as "not told" rather than discarding them.

3.3. Preprocessing and feature engineering

Each respondent is their own primary sampling unit, so there are no duplicate people or clusters for a random split to cut through. All preprocessing lives inside scikit-learn pipelines [23], so every statistic it uses is learned from the training split alone. Missing numeric values were filled with the training median, and a binary indicator was added for each feature that had gaps, so a model could still use the fact that an answer was missing. Continuous variables were capped at their training 0.1st and 99.9th percentiles. That limits the leverage of implausible BMI values without deleting the people who reported them. Smoking status was one-hot encoded with an explicit "missing" category. I standardised features for logistic regression and the neural networks, whose gradient-based training is sensitive to scale, but not for the tree ensembles, which are unaffected by it.

I also engineered six features from domain knowledge: a WHO BMI category, a count of diagnosed cardiometabolic conditions, a lifestyle-risk score, combined unhealthy days capped at 30, an age-by-BMI interaction and a flag for barriers to care. The transformer that builds them learns nothing from the data, so it cannot leak information between splits.

3.4. Splitting and evaluation design

A stratified 70/15/15 split gave 308,460 training, 66,099 validation and 66,099 test respondents. Validation data were used to compare experiments, stop network training and tune decision rules; the test set was opened once, at the end. Searches used three-fold cross-validation on a 100,000-row training sample, a compromise forced by my two-core CPU, and I also report survey-weighted test metrics as a rough population check.

With only 1.7% of respondents in the prediabetes class, accuracy says almost nothing, so I selected models on macro average precision, which averages the area under each class's precision–recall curve so that every class counts equally [17]. Because the models almost never chose a minority class as most probable, I tuned per-class probability multipliers on validation to maximise macro-F1, the three-class equivalent of moving a threshold. For a binary screening view I chose the diabetes cut-off maximising F_2, which values recall twice as much as precision.

3.5. Traditional machine learning

The traditional models were L2-penalised multinomial logistic regression, random forests [15] and gradient boosting [14], using scikit-learn's histogram-based implementation so that

boosting on 308,460 rows was feasible on a CPU; kernel SVMs, whose cost grows faster than linearly with sample size, were left out. The boosting search drew 16 candidates over learning rate, iterations, tree size, leaf size and L2 regularisation, and the forest search 10 candidates over depth, leaf size and features per split (the ranges are listed in the notebook, Section 8.3).

3.6. Deep learning

The neural networks were written in TensorFlow/Keras [24] and fed through `tf.data` pipelines. The training pipeline is cached, reshuffled every epoch, batched and prefetched, while validation and test pipelines keep a fixed order. Every network ends in a three-way softmax, trained with cross-entropy and Adam for up to 60 epochs, with early stopping after six epochs without improvement, checkpointing and TensorBoard logging.

I began with a Sequential network with one 32-unit hidden layer, to see whether any non-linearity helped, then a deeper one (128, 64 and 32 units with batch normalisation, 30% dropout and L2 decay of 10^{-4}). From there I changed one thing at a time: learning rate (10^{-2}, 10^{-3}, 10^{-4}), batch size (512 or 4096) and softened class weights. For the Functional API I built a wide-and-deep network [16], in which a linear branch, equivalent to logistic regression, and a regularised deep branch read the same inputs and are added before the softmax. I chose it because logistic regression had already captured most of the signal, so the deep branch only needs to learn what is left over.

3.7. Tracking, explanation and reproducibility

Every run was recorded with its own identifier and timestamp in a JSON log, a CSV results table and TensorBoard. All random seeds were fixed and TensorFlow's deterministic operations enabled, and running the whole notebook three times from scratch produced identical results. Explanations came from permutation importance, TreeSHAP [20] and standardised logistic regression coefficients, and test performance was broken down by sex, age, race and ethnicity, and income. Everything ran on a two-core CPU with 8.4 GB of RAM.

4. Results

4.1. What the data look like

Of the 440,658 labelled respondents, 85.26% reported no diabetes, 1.75% prediabetes and 12.99% diabetes. Survey weighting brings the diabetes share down to 10.5%, matching the codebook estimate. Income was the answer most often missing (17.97%), but its absence barely related to the outcome. Prevalence climbed steeply with age, from about 1% at 18–29 to about 20% at 70 and over, and with poorer self-rated health, from about 2% to 36%. Some strong associations reflected contact with health care rather than risk: respondents who had recently had their cholesterol checked were five times as likely to report diabetes (14.9% versus 2.8%).

4.2. How the experiments unfolded

Table 2 and Fig. 1 summarise all 17 experiments; Appendix A gives each run's settings and my observation on it.

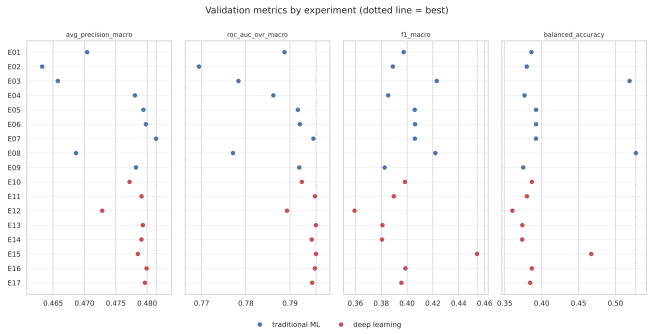


Figure 1: Validation metrics for all 17 experiments (blue: traditional ML; red: deep learning). Axes are zoomed to the observed range; the dotted line marks the best value.

Logistic regression set the baseline at a validation macro AP of 0.4704 (E01). Removing standardisation (E02) left the model unchanged but made it hard to fit: the solver ran out of iterations, took about five times longer and finished worse on every metric. A forest of fully grown trees (E04), averaging 38.9 levels and about 12,955 leaves, nearly memorised its samples: it improved average precision (0.4780) but ranked slightly worse by ROC-AUC, a classic sign of high variance. Gradient boosting (E05) improved on both measures. Adding the engineered features (E06) moved macro AP by only 0.0004, a reminder that trees already build interactions such as age and BMI for themselves.

Tuning was instructive in its own right. The boosting search preferred its most conservative candidate: a slow learning rate of 0.036 over 587 iterations, small 16-leaf trees and strong L2 regularisation. Its score on the training folds (0.625) stayed closest to its score on the held-out folds (0.475), while larger trees pushed training scores towards 0.80 without improving held-out performance at all. Refitted on the full training set (E07), this model achieved the best validation result of the study, a macro AP of 0.4814. The forest search likewise preferred shallow trees (E09), which lifted ROC-AUC from 0.7862 to 0.7921 without changing average precision.

The networks told a similar story. The 32-unit network (E10) stopped at epoch 9 with almost no train–validation gap (macro AP 0.4772), and the deeper regularised network (E11) reached 0.4791. A learning rate of 10^{-2} (E12) made training unstable, peaking at epoch 2, while a rate of 10^{-4} (E13) or batches of 4096 (E14) only slowed learning to the same point. The wide-and-deep network (E16) was the best deep model, with a macro AP of 0.4799. The learning curves (Fig. 2) show no overfitting: validation loss levels off rather than rising, and in the regularised networks training loss sits slightly above it because dropout only operates during training.

Class imbalance cut across every family. Choosing the most probable class, no model ever predicted prediabetes, and none found more than about a fifth of diabetes cases. Class weighting raised diabetes recall to 0.59 for logistic regression (E03) and 0.66 for boosting (E08), at the cost of accuracy near 0.64 and lower average precision; in the network (E15) it reached 0.50 with average precision unchanged. Weighting moved the decision boundary without improving the ranking.

Table 2: Experiment results on the validation set (66,099 respondents; argmax decisions). Split: stratified 70/15/15, seed 42. B = 25 base features, B+E = base + 6 engineered; CW = class weighting. Time = training seconds on 2 CPU cores. Macro AP is the selection metric; the best value is in bold. Full hyper-parameters are in Appendix A and `reports/experiment_log.jsonl`.

ID	Model	Feat.	Imb.	Time (s)	Acc.	Macro F1	Macro ROC-AUC	Macro AP	Rec. pre.	Rec. dia.
E01	LR baseline	B	–	5	0.858	0.397	0.7887	0.4704	0.00	0.18
E02	LR unscaled	B	–	26	0.856	0.389	0.7694	0.4632	0.00	0.16
E03	LR balanced	B	CW	5	0.645	0.423	0.7783	0.4657	0.31	0.59
E04	RF baseline	B	–	26	0.860	0.385	0.7862	0.4780	0.00	0.14
E05	GB baseline	B	–	26	0.860	0.406	0.7918	0.4794	0.00	0.20
E06	GB + engineered	B+E	–	33	0.861	0.406	0.7922	0.4798	0.00	0.20
E07	GB tuned	B+E	–	84	0.861	0.406	0.7953	**0.4814**	0.00	0.20
E08	GB tuned, balanced	B+E	CW	84	0.638	0.422	0.7771	0.4687	0.28	0.66
E09	RF tuned	B+E	–	53	0.860	0.383	0.7921	0.4782	0.00	0.14
E10	MLP 32 (Seq.)	B	–	28	0.859	0.398	0.7927	0.4772	0.00	0.18
E11	MLP 128-64-32 reg. (Seq.)	B	–	112	0.860	0.390	0.7956	0.4791	0.00	0.16
E12	E11, lr 10^{-2}	B	–	33	0.858	0.359	0.7893	0.4728	0.00	0.09
E13	E11, lr 10^{-4}	B	–	216	0.860	0.381	0.7958	0.4793	0.00	0.13
E14	E11, batch 4096	B	–	130	0.860	0.381	0.7949	0.4791	0.00	0.13
E15	E11, class-weighted	B	CW	78	0.834	0.454	0.7959	0.4785	0.00	0.50
E16	Wide-and-deep (Func.)	B	–	106	0.860	0.399	0.7956	0.4799	0.00	0.18
E17	E16 + engineered	B+E	–	104	0.860	0.396	0.7950	0.4797	0.00	0.17

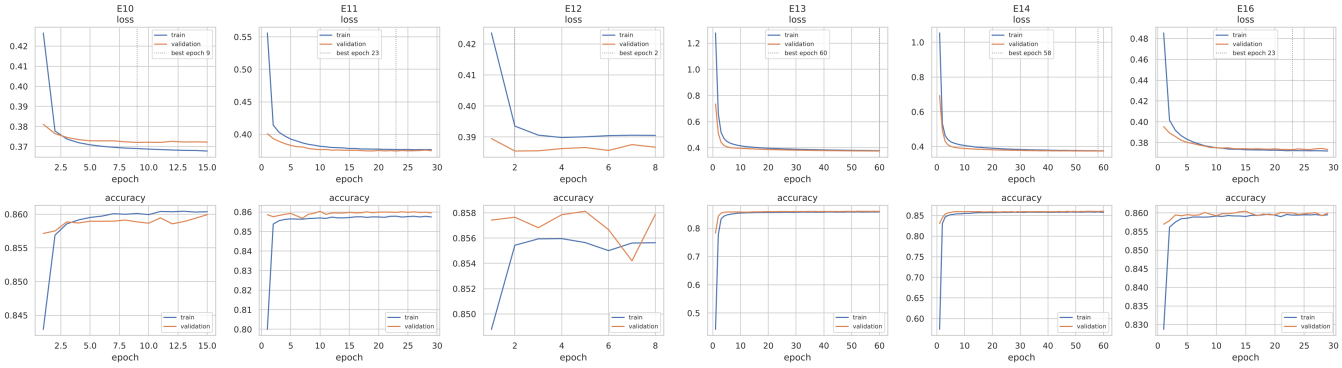


Figure 2: Training and validation loss (top) and accuracy (bottom) for six Keras experiments. The dotted line marks the early-stopping epoch.

4.3. The final test

I carried tuned boosting (E07) and the wide-and-deep network (E16) forward, with logistic regression (E01) as an interpretable reference. Tuning each model's class multipliers on validation ({1, 15, 2.5} for E07 and E16, {1, 20, 3} for E01) lifted validation macro-F1 from about 0.40 to 0.47. On the test set (Table 3), boosting and the network tied on macro ROC-AUC (0.8004 and 0.8005), boosting kept a small lead in average precision (0.4790 against 0.4775), and logistic regression followed about 0.01 behind; test scores tracked validation closely, so selection had not overfitted.

The confusion matrix in Fig. 3 shows where the errors fall. Boosting found 4,296 of the 8,588 people with diabetes, and four in ten of its diabetes flags were correct. Most of the cases it missed (3,987) were labelled as having no diabetes, while 5,549 people without diabetes were flagged in the other direction. Prediabetes was hardly recognised: 54 of 1,154 cases were identified, with 714 assigned to no diabetes and 386 to diabetes. The ROC and precision–recall curves in Fig. 4 make the contrast between the two minority classes stark. Diabetes reached a ROC-AUC of 0.843 and an average precision of 0.434, more than three times its 13% base rate.

Table 3: Final held-out test results (66,099 respondents) with validation-tuned decision weights. E01 = logistic regression, E07 = tuned gradient boosting, E16 = Functional wide-and-deep network. Best value per row in bold.

Metric	**E01**	**E07**	**E16**
Accuracy	0.805	0.823	**0.824**
Balanced accuracy	**0.480**	0.478	0.475
Macro F1	0.462	**0.468**	0.466
Macro ROC-AUC	0.794	0.800	**0.800**
Macro AP	0.469	**0.479**	0.477
Recall, prediabetes	**0.072**	0.047	0.042
Precision, prediabetes	0.044	**0.048**	0.047
Recall, diabetes	**0.502**	0.500	0.492
Precision, diabetes	0.394	**0.420**	0.418
Macro F1 (survey-weighted)	0.478	**0.480**	0.478
Macro ROC-AUC (survey-weighted)	0.823	**0.828**	0.826
Recall, diabetes (survey-weighted)	0.469	**0.479**	0.465

Prediabetes reached a respectable-looking ROC-AUC of 0.72, but its average precision of 0.036 barely cleared its 1.7% base rate.

4.4. Screening, errors, explanations and subgroups

Viewed as a screening tool (Table 4), the model's usefulness depends almost entirely on its threshold. At the default cut-off

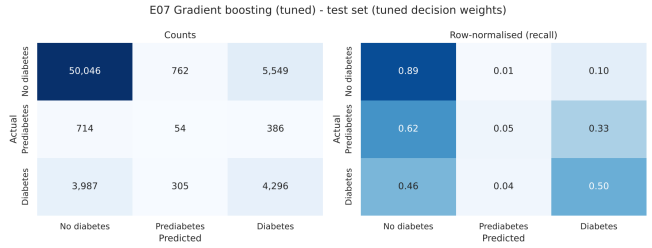


Figure 3: Test-set confusion matrices for tuned gradient boosting (E07) with tuned decision weights: counts (left) and row-normalised recall (right).

Table 4: Binary screening view (diabetes vs. not) for tuned gradient boosting on the test set. The F_2 threshold was selected on validation.

Rule	Thr.	Sens.	Spec.	PPV	NPV	Flagged
Default 0.5	0.500	0.168	0.982	0.581	0.888	0.038
F_2-optimal	0.112	0.847	0.680	0.283	0.967	0.389
F_2-optimal, weighted	0.112	0.809	0.764	0.287	0.971	0.296

of 0.5, boosting caught only 16.8% of diabetes cases. At the F_2 threshold of 0.112 chosen on validation, it caught 84.7% and a negative result was right 96.7% of the time, but it flagged 38.9% of all respondents, and only 28.3% of those flagged actually had diabetes. The cases it missed looked more like the general population than like the cases it found, with an average BMI of 28.8 against 34.4 for detected cases and fewer diagnosed conditions, while the false alarms came disproportionately from people with high-risk profiles.

The explanation methods agreed on the core signals. Permutation importance put BMI first (shuffling it cost 0.0347 in macro AP), then self-rated health, age, hypertension and high cholesterol. SHAP (Fig. 5) placed self-rated health first and credited the engineered comorbidity count and age-by-BMI interaction, and logistic regression gave odds ratios per standard deviation of 1.73 for poorer self-rated health, 1.56 for age and 1.50 for BMI. Some signals, though, ran in directions no causal account would predict. A recent check-up and a cholesterol test raised the predicted risk, while heavy drinking lowered it.

Finally, Table 5 shows similar performance for men and women but not across age and income. Among adults under 45 the model ranked best (ROC-AUC 0.877) yet caught only 20.9% of cases, while among those 65 and over it caught 54.4% but ranked worst (0.767). Recall rose from 0.370 in the highest income band to 0.631 in the lowest. Among racial and ethnic groups, Black non-Hispanic respondents had the highest prevalence (20.5%) and the lowest ROC-AUC (0.813).

5. Discussion

5.1. Why the models ended up so close

My hypothesis held in direction, but the effect turned out smaller than I expected. Tuned gradient boosting gave the best threshold-free performance, a well-regularised network matched it on ROC-AUC, and logistic regression was only about 0.01 behind in average precision. This agrees with the clinical review of Christodoulou *et al.* [8] and with the tabular benchmarks [6, 7], and my experiments suggest a reason. In these data, most of the signal is additive. The best network was the one containing a logistic regression, engineered interactions helped neither paradigm, and every network landed between 0.473 and 0.480 macro AP whatever its depth or learning rate.

Table 5: Test-set diabetes detection by subgroup (tuned gradient boosting, tuned decision weights). Race/ethnicity was not a model input. All subgroups have at least 163 diabetes cases.

Dim.	Group	n	Prev.	Recall	Prec.	AUC
sex	Female	37,957	0.124	0.486	0.417	0.846
sex	Male	28,142	0.138	0.517	0.423	0.839
age	45-64	25,676	0.131	0.485	0.418	0.832
age	65+	22,944	0.205	0.544	0.421	0.767
age	18-44	17,479	0.030	0.209	0.429	0.877
race/ethnicity	White NH	50,483	0.121	0.494	0.404	0.840
race/ethnicity	Hispanic	5,223	0.145	0.518	0.474	0.866
race/ethnicity	Black NH	5,127	0.205	0.543	0.462	0.813
race/ethnicity	Other NH	2,946	0.125	0.447	0.498	0.869
race/ethnicity	Multiracial NH	1,241	0.131	0.540	0.440	0.870
income	\$50k+	26,238	0.090	0.370	0.413	0.856
income	\$20-50k	18,562	0.152	0.532	0.421	0.825
income	<\$20k	9,620	0.199	0.631	0.438	0.816

The tuning results show where useful complexity runs out. Giving the models more capacity, whether deeper forests or larger boosted trees, raised their scores on training folds by as much as 0.18 without improving held-out performance, and in both searches the winning configuration was the most heavily regularised one on offer. The boosting learning curve tells the same story. Training and validation loss settled with a small, steady gap (0.357 against 0.370) and the validation curve never turned upwards, so the model sits close to the point where extra variance stops buying lower bias. My diabetes AUC of 0.843 is also in line with the 0.82–0.83 that Chowdhury *et al.* [12] found on the binary derivative, close to their estimated ceiling. Taken together, this suggests that the limit lies in what the survey measures rather than in the algorithms, and it adds to my doubts about reported AUCs near 0.99 on the same variables [10].

5.2. Thresholds matter more than algorithms

The single most consequential decision in this project was the decision rule, not the choice of model. With default decisions, every model was more than 85% accurate while finding at most a fifth of diabetes cases and no prediabetes cases at all, a textbook case of accuracy misleading under imbalance [17]. Both class weighting and tuned decision multipliers improved minority-class recall, but weighting did so at the cost of average precision, whereas adjusting the multipliers after training left the probability estimates untouched. I prefer the second approach because it keeps ranking quality separate from where the line is drawn, and the F_2 threshold shows what drawing that line entails. It found 85% of diabetes cases, but only by flagging almost two in five respondents. Weighting recall more heavily makes sense when a missed case means an untreated disease and a false alarm means a confirmatory blood test. Even so, I would not recommend any clinical threshold on the strength of these results alone.

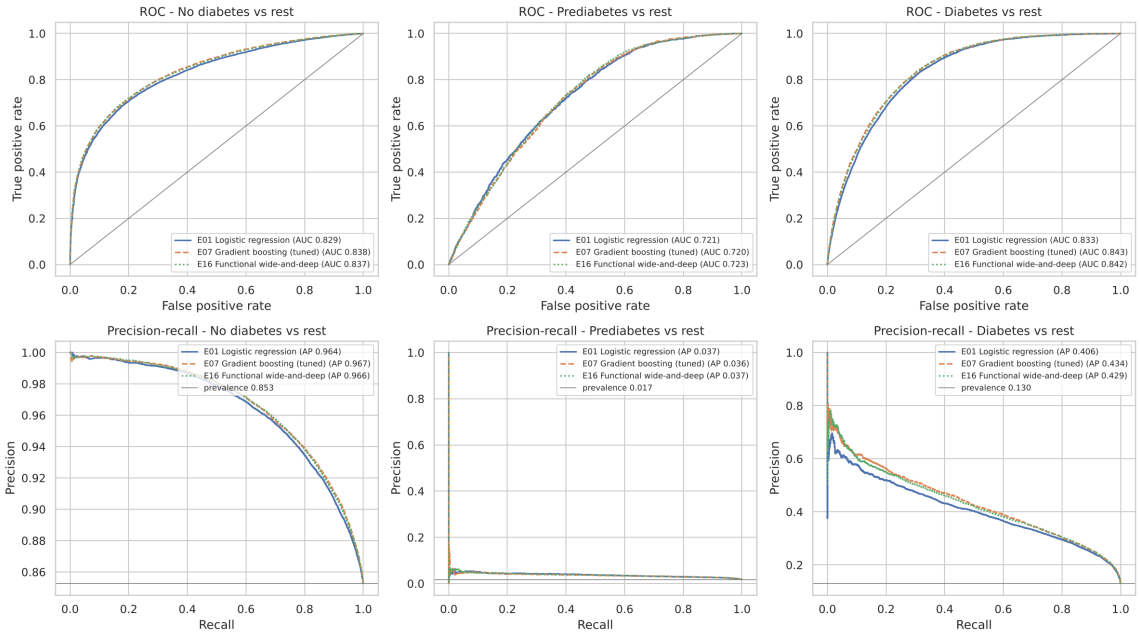


Figure 4: One-vs-rest ROC (top) and precision–recall (bottom) curves on the test set for the three final models. Grey lines show chance and class prevalence.

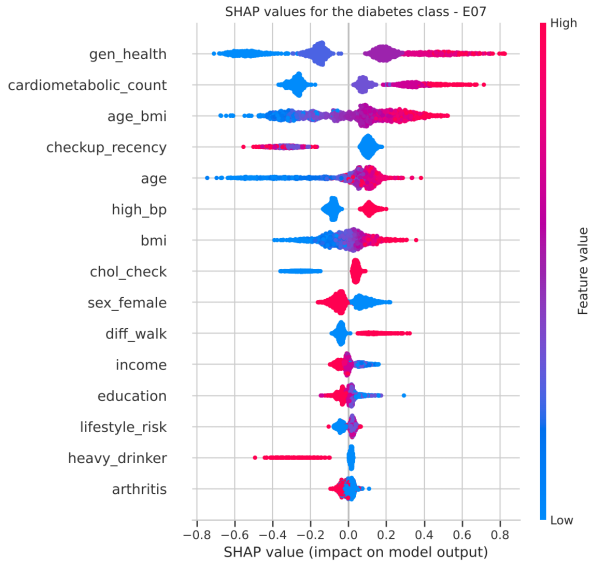


Figure 5: TreeSHAP values for the diabetes class (E07, 2,000 validation respondents). Colour shows the feature value; positive SHAP values push predictions towards diabetes.

5.3. *What the errors reveal about the labels*

The error analysis changed how I see the false positives. Many wrongly flagged people look, on paper, like people with diabetes, and because the label records whether someone was *told*, undiagnosed cases count as negatives; some of these "errors" may be correct. Prediabetes suffers most, since it only appears in the data if someone happened to be tested and informed. That probably explains why it occupied a blurred middle ground between its neighbours, and why its average precision of 0.036 says as much about the label as about the models. The data also blur cause and effect. Poorer self-rated health, a recent check-up, a cholesterol test and drinking less are all things that become more likely *after* a diagnosis, and Xie *et al.* [9] saw the same check-up pattern. The feature importances therefore describe what a classifier can exploit in cross-sectional data, not what causes diabetes [19].

5.4. *Representativeness and fairness*

The BRFSS is large and nationally weighted, but its self-reported prevalence differs from surveys with physical measurements [22]. Survey-weighted test metrics stayed close to the unweighted ones (weighted macro ROC-AUC 0.828), so my conclusions do not appear to be an artefact of the sample's age structure. The subgroup results are more troubling: one threshold gave recall of 0.21 for adults under 45 against 0.63 in the lowest income band, and the groups with the most diabetes were the hardest to rank. Because getting a diagnosis depends on access to care, the quality of the labels may itself differ between groups [21]. These disparities are descriptive and do not prove that the model is unfair, but they would need to be addressed before any real use.

5.5. *What this means for rural sub-Saharan Africa*

Three findings bear directly on my mission. First, a simple model is enough: logistic regression came within about 0.01 of the best, runs on any phone, and has coefficients a community health worker can understand. Second, choosing a threshold is a decision about resources. Catching 85% of cases meant flagging 39% of people, which is only sensible where clinics can supply confirmatory tests on that scale. Third, and most important, "ever told" labels undercount the people with least access to care, which describes the rural communities where most diabetes goes undiagnosed [2,3]; a model trained on such labels would learn to miss the very people it is meant to find. Any African deployment would have to begin by collecting laboratory-confirmed labels locally and revalidating the model

on them.

5.6. Limitations and next steps

The searches also used a 100,000-row subsample, the data cover only US adults in 2015, and I did not assess calibration. The most valuable next steps are to validate against laboratory-defined diabetes, for instance HbA1c measurements in NHANES, so that undiagnosed cases are labelled correctly; to use design-based estimation with the survey strata; to add calibration checks and group-specific thresholds; and to test whether the models hold up on later survey years.

6. Conclusion

I set out to learn how well traditional machine learning and deep learning can classify diabetes status from population health indicators, and how the two approaches differ. Working with 440,658 respondents to the raw 2015 BRFSS, keeping prediabetes as its own class and imputing missing answers rather than discarding them, I found that both approaches reach a similar and fairly modest ceiling. On the held-out test set, tuned gradient boosting achieved a macro ROC-AUC of 0.800 and a macro average precision of 0.479, a wide-and-deep Keras network matched its ROC-AUC, and logistic regression came within about 0.01 while training in seconds and remaining fully interpretable. Diabetes itself was identified reasonably well (ROC-AUC 0.843), but self-reported prediabetes was almost indistinguishable from its neighbouring classes.

The comparison is fully reproducible, from a codebook-verified data file to identical results on repeated runs. The decision rule, far more than the algorithm, determined whether a model found one diabetes case in six or five in six, and the error and subgroup analyses exposed label gaps, reverse causation and unequal sensitivity that headline accuracy would have hidden. The main limitations are the self-reported labels, the cross-sectional design, evaluation that does not fully account for the survey design, and the use of a single year of data.

For my own path, the most important lesson is that questionnaire-based screening can help, but only if the labels it learns from are trustworthy. The next step towards my mission is to validate models like these against laboratory-confirmed glycaemic status, to calibrate them and set thresholds group by group, and to test them on locally collected data from sub-Saharan African communities before considering any use in rural screening.

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A. Experiment configurations

Every row below corresponds to a run in `reports/experiment_log.jsonl` and a TensorBoard directory `logs/tensorboard/<run ID>`. All runs used the same stratified 70/15/15 split (seed 42). Traditional models used median imputation with missing-value indicators, 0.1/99.9% winsorising of continuous variables and one-hot encoded smoking status; logistic regression (except E02) and all networks were also standardised. Run IDs are from the final reported execution.

ID	Run ID	Hyper-parameters	Observation (after reading the results)
E01	`E01_20261018T073349Z`	C=1.0, penalty=l2, solver=lbfgs, max_iter=1000	Reference model: converges in 36 iterations; 0.86 accuracy hides 0 prediabetes and 0.18 diabetes recall under argmax.
E02	`E02_20261018T073416Z`	C=1.0, penalty=l2, solver=lbfgs, max_iter=200, scaling=False	Without scaling, L-BFGS hits its 200-iteration cap (convergence warning), runs 5x slower and is worse than E01 on every metric.
E03	`E03_20261018T073421Z`	C=1.0, penalty=l2, solver=lbfgs, max_iter=1000	Balanced weights vs E01: diabetes recall 0.18->0.59, prediabetes 0->0.31, but accuracy 0.645 and macro AP drops slightly; threshold moved, ranking not improved.
E04	`E04_20261018T073449Z`	n_estimators=150, max_depth=None, min_samples_leaf=5, max_features=sqrt	Fully grown trees (mean depth 39, 13k leaves): macro AP above E01 but ROC-AUC below it, the signature of a high-variance learner.
E05	`E05_20261018T073517Z`	learning_rate=0.1, max_iter=200, max_leaf_nodes=31, min_samples_leaf=20	Boosting improves both macro AP and ROC-AUC over the forest: sequential bias reduction beats averaging deep trees here.
E06	`E06_20261018T073551Z`	learning_rate=0.1, max_iter=200, max_leaf_nodes=31, min_samples_leaf=20	Engineered features add only +0.0004 macro AP to E05; trees already form interactions such as age x BMI by splitting.
E07	`E07_20261018T075447Z`	l2_regularization=4.33792, learning_rate=0.03629, max_iter=587, max_leaf_nodes=16, min_samples_leaf=73	Search picks the most regularised candidate (lr 0.036, 16 leaves, L2 4.34; train-fold 0.625 vs CV 0.475): best macro AP of all runs.
E08	`E08_20261018T075614Z`	l2_regularization=4.33792, learning_rate=0.03629, max_iter=587, max_leaf_nodes=16, min_samples_leaf=73	Balanced weights on E07 repeat the E03 pattern: diabetes recall 0.66, prediabetes 0.28, macro AP lower.
E09	`E09_20261018T080006Z`	max_depth=14, max_features=0.1572, min_samples_leaf=6, n_estimators=300	Shallower forest (depth 14, 16% features per split) raises ROC-AUC 0.786->0.792 at unchanged macro AP: regularisation smooths mid-risk ranking.
E10	`E10_20261018T080040Z`	layers=32, activation=relu, learning_rate=0.001, dropout=0, l2=0, batch_size=512, epochs_run=15, best_epoch=9, params=1763, optimizer=Adam, early_stopping=val_loss, patience 6	32-unit MLP early-stops at epoch 9 with a negligible train/val gap; beats LR slightly but trails boosting.
E11	`E11_20261018T080233Z`	layers=128-64-32, batch_norm=True, dropout=0.3, l2=0.0001, learning_rate=0.001, batch_size=512, epochs_run=29, best_epoch=23, params=17987, optimizer=Adam, early_stopping=val_loss, patience 6	Deeper MLP with BN/dropout/L2: small gain over E10; training loss above validation loss because regularisation is active only in training.
E12	`E12_20261018T080307Z`	layers=128-64-32, batch_norm=True, dropout=0.3, l2=0.0001, learning_rate=0.01, batch_size=512, epochs_run=8, best_epoch=2, params=17987, optimizer=Adam, early_stopping=val_loss, patience 6	lr 1e-2 is unstable (best epoch 2, oscillating validation loss) and loses macro AP versus E11.
E13	`E13_20261018T080645Z`	layers=128-64-32, batch_norm=True, dropout=0.3, l2=0.0001, learning_rate=0.0001, batch_size=512, epochs_run=60, best_epoch=60, params=17987, optimizer=Adam, early_stopping=val_loss, patience 6	lr 1e-4 is still improving at the 60-epoch cap and takes 2x E11's time for the same macro AP: too slow for the budget.
E14	`E14_20261018T080856Z`	layers=128-64-32, batch_norm=True, dropout=0.3, l2=0.0001, learning_rate=0.001, batch_size=4096, epochs_run=60, best_epoch=58, params=17987, optimizer=Adam, early_stopping=val_loss, patience 6	Batch 4096 needs 58 of 60 epochs to match E11: fewer updates per epoch, no quality gain.
E15	`E15_20261018T081015Z`	layers=128-64-32, batch_norm=True, dropout=0.3, l2=0.0001, learning_rate=0.001, batch_size=512, epochs_run=19, best_epoch=13, params=17987, optimizer=Adam, early_stopping=val_loss, patience 6	Softened class weights raise diabetes recall 0.16->0.50 and macro-F1 to 0.454 with macro AP unchanged; prediabetes still never predicted.
E16	`E16_20261018T081202Z`	architecture=wide (linear) + deep 128-64, logits summed, dropout=0.3, l2=0.0001, learning_rate=0.001, batch_size=512, epochs_run=29, best_epoch=23, params=16031, optimizer=Adam, early_stopping=val_loss, patience 6	Wide-and-deep is the best deep model: the linear wide path keeps the additive signal, the deep path adds little on top.
E17	`E17_20261018T081347Z`	architecture=wide (linear) + deep 128-64, logits summed, dropout=0.3, l2=0.0001, learning_rate=0.001, batch_size=512, epochs_run=29, best_epoch=23, params=17603, optimizer=Adam, early_stopping=val_loss, patience 6	Engineered features give no gain on E16, consistent with E06.

B. Supplementary figures

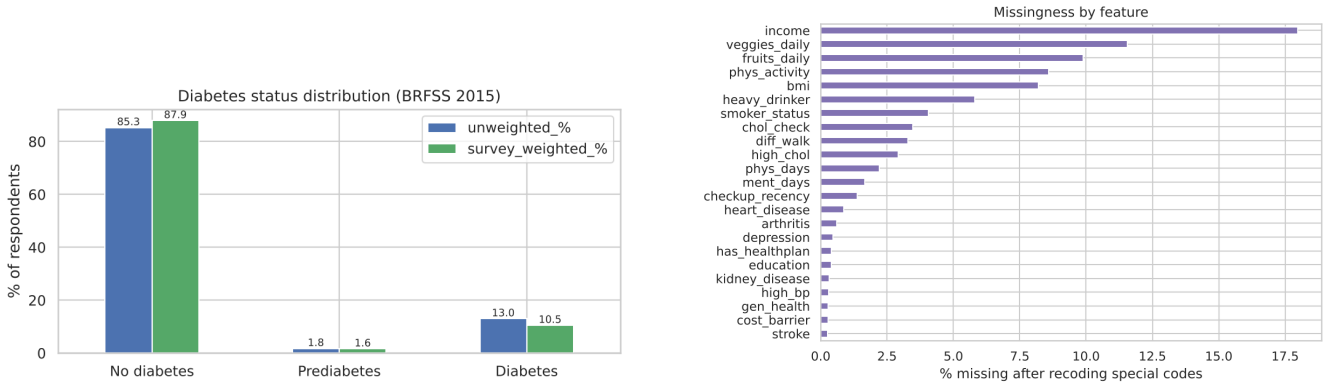


Figure 6: Left: class distribution, unweighted and survey-weighted. Right: percentage of missing values per feature after recoding special codes.

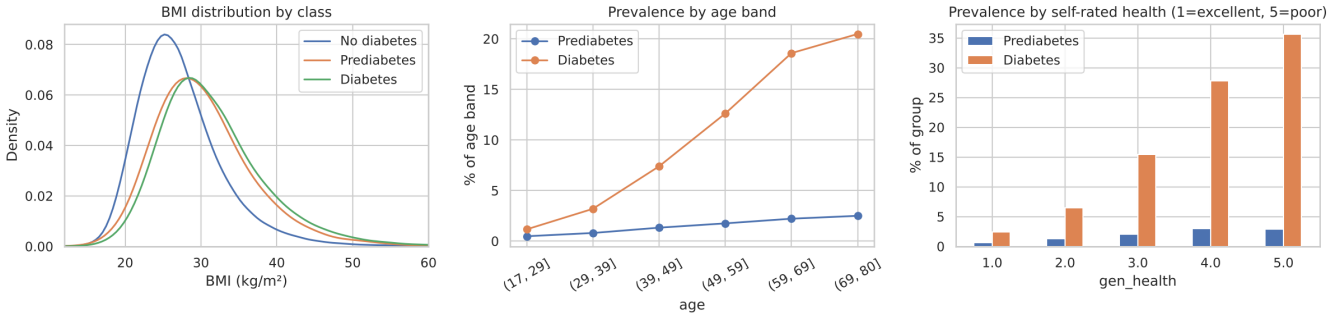


Figure 7: BMI distribution by class, prevalence by age band, and prevalence by self-rated general health.

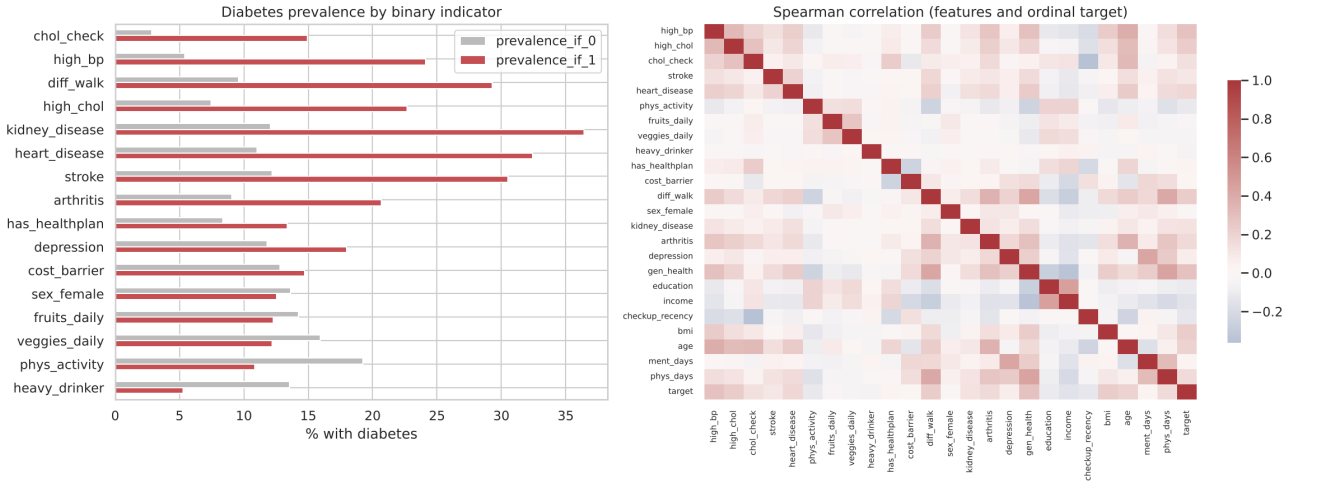


Figure 8: Diabetes prevalence by binary indicator (left) and Spearman correlations among features and the ordinal target (right).

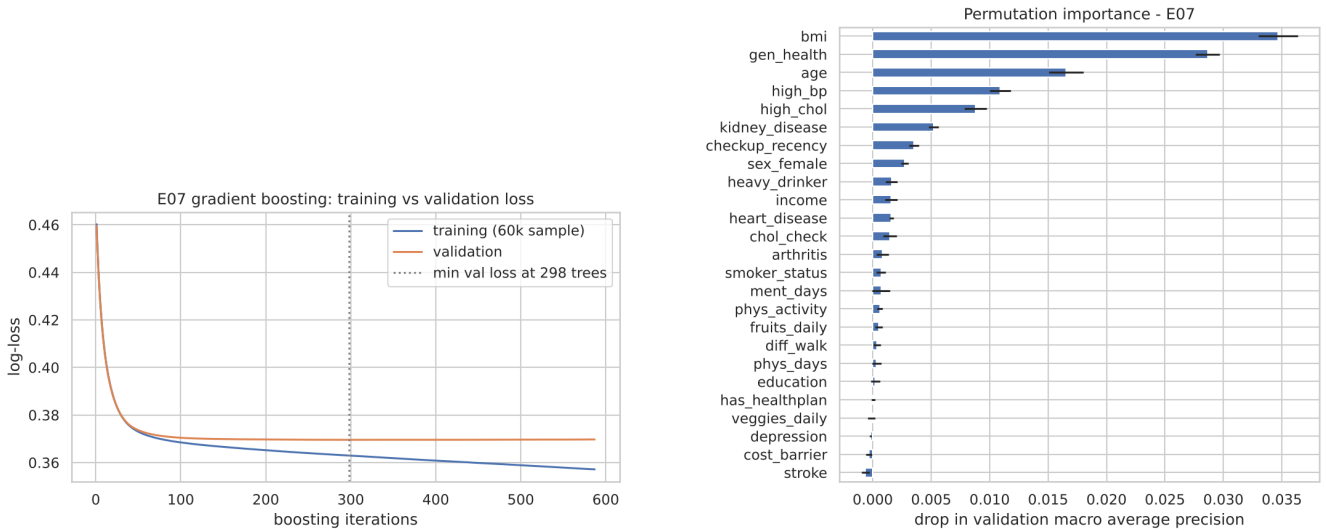


Figure 9: Left: training vs. validation log-loss across boosting iterations for E07. Right: permutation importance (drop in validation macro AP) for E07.

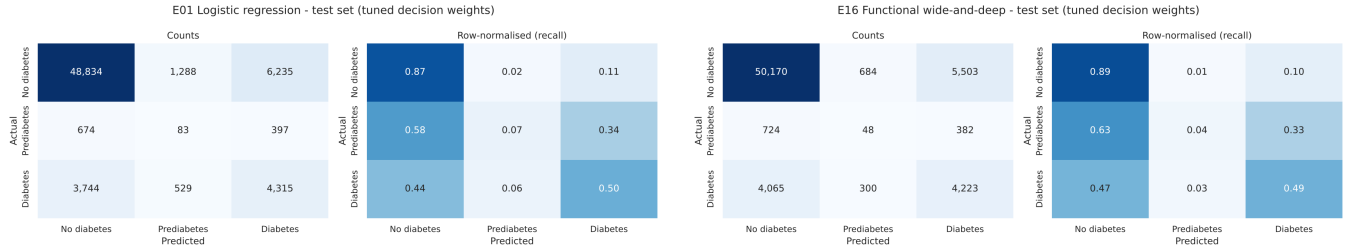


Figure 10: Test-set confusion matrices for logistic regression (E01, left) and the wide-and-deep network (E16, right) with tuned decision weights.

C. Reproducibility

The environment (Python 3.13, scikit-learn 1.9.1, TensorFlow 2.21, Keras 3.15, SHAP 0.54) is pinned in `requirements.txt` and recorded in `reports/environment.json`. The SHA-256 of the data file used is `bfe9e629...ad300b1b9`. Saved artifacts (full scikit-learn pipeline, Keras model with its preprocessor, and metadata containing decision weights and the screening threshold) were reloaded and reproduced the in-memory predictions exactly.